

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>
Here’s a concise solution outline:

- Kinetic energy: $E = 1 \text{ eV} \approx 1.602 \times 10^{-12} \text{ erg}$.
- Relate kinetic energy to momentum using $E = p^2 / (2m)$, solving for p gives $p = \sqrt{2mE}$.
- Mass of electron: $m_e \approx 9.109 \times 10^{-28} \text{ g}$.
- Compute momentum: $p = \sqrt{2 \times (9.109 \times 10^{-28} \text{ g}) \times (1.602 \times 10^{-12} \text{ erg})} \approx 4.36 \times 10^{-21} \text{ g}\cdot\text{cm/s}$.
- De Broglie wavelength: $\lambda = h/p$.
- Planck’s constant in the appropriate units for cm: $h \approx 6.625 \times 10^{-27} \text{ erg}\cdot\text{sec}$.
- Thus $\lambda \approx (6.625 \times 10^{-27}) / (4.36 \times 10^{-21}) \approx 1.52 \times 10^{-6} \text{ cm} \approx 1.23 \times 10^{-7} \text{ cm}$.

Therefore